

First Semester

Name of Paper: English Language & Communication Skills

Course No.: BCA/1/CC/01

Marks Scale: **100** marks (End Sem. Exam: 75+Int.: 25)

Credit: **4**
(3- 0- 1)

Unit 1: Grammar 1

(10L)

Parts of Speech, Definition & Identification of Subject and Predicate, Phrases & Clauses, Tense – Types of Tenses & their use.

Unit 2: Grammar 2

(10L)

Voice – Active voice and Passive voice, Concept of Concord – What is Concord? Subject - Verb Agreement; Reported Speech – Direct and Indirect Speech.

Unit 3: Introduction to Language Communication

(10L)

Importance of English Language, Basics of Communication – Process of Communication, Components of Communication, factors of Communication; Barriers to Communication – Physical, Psychological, Semantics, Organizational and Interpersonal Barriers; How to overcome Barriers.

Unit 4: Communication Skills in English

(10L)

Language Skills- Reading Skills and Listening Skills; Verbal Communication- Vocal Communication techniques and Oral Presentation; Non Verbal Communication- Personal appearance; Facial Expression, Movement, Posture, Gesture, Eye Contact.

Unit 5: Oral Communication

(P = 1Credit)

(Practical to be conducted in the College)

Facing Interview-Viva Voce, Different forms of classroom interaction-seminar, paper presentation, Group Discussion, Public Speaking.

Recommended Books:

- 1) R.C.Sharma& Krishna Mohan: *Business Correspondence & Report Writing, A Practical Approach to Business and Technical Communication*, Tata McGraw Hill. **(New)**
- 2) Aruna Koneru: *Professional Communication*, Tata McGraw Hill Pub. Co. Ltd.
- 3) Wren & Martin: *English Grammar and Composition*, S. Chand Publisher (2011)

First Semester
Name of Paper: Basic Mathematics
Course No.: BCA/1/CC/02

Marks Scale: 100 marks (End Sem. Exam: 75+Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

(10L)

Number system: Natural number, whole number, integer, real, even, odd, prime, composite, rational, irrational, decimal numbers; Test for divisibility of number; LCM and HCF of numbers; Square and cube roots; Surds and indices; **Fractions:** Insertion of any number between two given fraction; Simplification(VBODMAS); Algebraic formula

Unit-2:

(10L)

Algebraic expression: monomial, binomial, trinomial, polynomial, degree of polynomial, multiplication and division of polynomial; Factorisation; Linear equations in one and two variables; Quadratic equation; Problems on numbers; Ratio and proportion; Average; Percentage

Unit-3:

(10L)

Profit and loss; simple interest; compound interest; Discount; Partnership; Alligation (or mixture); Stock and shares; Time, Work and Distance; Pipes and Cistern; Boats and stream; Problems on train; Calendar and clocks; Data interpretations

Unit-4:

(10L)

Sequence and series: Definition of sequence, series and progression; Arithmetic progression(AP): n^{th} term of an AP, Sum of n terms of an AP, Arithmetic mean(AM), Properties of an AP; Geometric progression(GP): n^{th} term of GP, Sum of n terms of GP, Geometric mean(GM), Properties of GP; Harmonic progression(HP); Harmonic mean(HM); Relation between AM, GM and HM

Unit-5:

(10L)

Matrices: Definition of a matrix; Types of matrices, Addition and multiplication of matrices; Transpose of matrix: symmetric and skew symmetric matrices; Row and column operations; inverse of a matrix by using elementary row operations; Adjoint and inverse of a matrix; System of linear equations by matrix method; Determinants: minors and cofactors; expansion of determinants; properties of determinants.

Recommended Books:

1. R.S. Aggarwal: Senior Secondary School Mathematics for Class 11, Bharti Bhavan (Publishers and Distributors)
2. R.S. Aggarwal: Senior Secondary School Mathematics for Class 12, Bharti Bhavan (Publishers and Distributors)
3. SSC CGL Pre Examination, Compiled and edited by Arihant Expert Team, Arihant Publication (India) Limited
4. Quantitative Aptitude by Dr R.S. Aggarwal, S. Chand Publication

First Semester

Name of Paper: Introduction to Information Technology

Course No.: BCA/1/CC/03

Marks Scale: **100** marks (End Sem. Exam: 75+Int.: 25)

Credit: 4

(3- 1- 0)

Unit –1:

(10 L)

Data and Information: Introduction, Types of Data, A Simple Model of a Computer, Data Processing Using a Computer, Desktop Computer, Data storage, storage cell, physical device used as memory cells, random access memory, read only memory, secondary storage, compact disk read only memory(CDROM), archival store

Central Processing Memory: Introduction, The Structure of a Central Processing Unit, Specification of a CPU, Interconnection of CPU with Memory and I/O Units, Embedded Processors

Unit –2:

(10 L)

Output devices: Introduction, Video Display Devices, Flat Panel Displays, Printers, Audio Output

Computer Software: Introduction, Operating System- Definition, functions of operating system, concept of multiprogramming, multitasking, multithreading, multiprocessing, time-sharing, real time, single-user & multi-user operating system, Programming Languages, A Classification of Programming Languages, assemblers, compilers, interpreters, linkers, algorithms, flow charting, decision tables, pseudo code.

Unit –3:

(10 L)

Software concepts: Definition, types of software: system & application software packages, hardware & software.

Data Organisation: Introduction, Organizing a Database, Structure of a Database, Database Management System, Example of Database Design, Non-text Databases, Archiving Databases.

Unit – 4:

(10 L)

Computer Networks: Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media, Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), OSI & TCP/IP Model, Network, Client and Servers, Host & Terminals, TCP/IP.

Unit – 5:

(10 L)

Some Internet Applications: Introduction, E-mail, Information Browsing Service, The World Wide Web, Information Retrieval from the World Wide Web, Other Facilities Provided by Browsers, Audio on the Internet, Pictures, Animation and Video via Internet

Societal Impacts of Information Technology: Introduction, Privacy, Security and Integrity of Information, Disaster Recovery, Intellectual Property Rights, Careers in Information Technology.

Recommended Books:

1. V. Rajaraman: *Introduction of Information Technology*, 2nd Edition, PHI Learning Private Limited (2013)
2. P. K. Sinha & Priti Sinha : *Computer Fundamentals*, BPB Publications (2009).
3. Peter Norton : *Introduction to Computers*, 4th Edition, TMH (2001).
4. Alex Leon & Mathews Leon : *Introduction to Computers*, Vikas Publishing House (2001)

First Semester
Name of Paper: Digital Computer Fundamentals
Course No.: BCA/1/CC/04

Marks Scale: **100** marks (End Sem. Exam: 75+Int.: 25)

Credit: **4**
(3- 1- 0)

Unit 1:

(10L)

Introduction to Number system and Codes – Digital computers and Digital Systems, Binary Numbers, Octal and Hexadecimal numbers, Number Base Conversion- different number systems and their conversions (Decimal, Binary, Octal, Hexadecimal), Complements- r's complement and (r-1)'s complement, Binary Codes- Decimal Codes, Error Detection Codes, Alphanumeric Codes.

Unit 2:

(10L)

Boolean algebra: Basic Definition, Basic theorems and properties of Boolean algebra, De-Morgan's theorems, Simplification of Boolean algebra, Boolean functions, Canonical and Standard forms, sum of minterm, product of maxterm, Karnaugh map- two, three and four variable maps, Sum of Product and Product of Sum simplification.

Unit 3:

(10L)

Digital Arithmetic: addition using 9's and 10's, addition using 1's and 2's compliment, multiplication and division of binary number.

Logic gates: Digital logic gates- AND, OR, NOT, XOR, XNOR, NAND, NOR. Universal property of NAND gate and NOR gate, implementation of Boolean function using logic gates.

Unit 4:

(10L)

Combinational Logic: Introduction, Design procedure, circuit diagram, truth table of Adders (half and full), Subtractor (half and full), Decoder (2:4, 3:8), Encoder (4:2, 8:3), Multiplexer (4:1, 8:1), De-multiplexer (1:4, 1:8), Programmable Logic Array (PLA).

Unit 5:

(10L)

Sequential Logic: Flip-Flops - RS flip-flops, D flip-flops, JK flip-flops. T flip-flops, Master slave flip-flop, edge trigger flipflop.

Registers: Register, Shift Register.

Asynchronous counters: Four bit ripple counter, BCD counter.

Synchronous counter: Binary Counter, BCD Counter.

Recommended Books:

1. M.Morris Mano: Digital Logic and Computer Design, Eastern Economy Edition, Prentice Hall of India, 2003.
2. Malvino, Brown: Digital Computer Electronics, 3rd Edition, 1995, Tata McGraw Hill.
3. Malvino, Paul Albert and Leach, Donald P: Digital Principles and Applications, 4th Edition, 2000. TMH.
4. Malvino, Paul Albert and Leach, Donald P: Digital Computer Fundamentals, 3rd Edition, 1995. TMH.

First Semester

Name of Paper: PC Applications & Internet Technology Lab

Course No.: BCA/1/CC/05

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: 3

(0- 0- 3)

A. Record Book

(10 marks)

B. Viva Voce

(15 marks)

C. Practical

(50 marks)

Windows Basic and DOS:

Personalizing Windows- Backgrounds, Themes, Working with Files and Folders, Windows Accessories - Paint, Notepad, Wordpad, etc. Windows Tools - Disk Cleanup, Defragmenter, System Restore, etc.

Important DOS commands, Internal Commands - CD, Cls, Color, Copy, Date, Del, Dir, Echo, Exit, Goto, Md, Move, Pause, Prompt, Rd, Ren, Time, Ver, Vol; External Commands - Append, At, Attrib, Chkdsk, Edit, Help, Ipconfig, Label, More, Netstat, Shutdown and creating Batch Files.

HTML

HTML 5 Documents Basics, DOCTYPE, Head, Body, and Body, CSS Styles, Headings and Heading Styles, Page Divisions: div for Structure and Layout, Paragraph and Text Styles, Links and Link Styles, Multimedia, Images, and Image Styles, Lists and List Styles, Tables and Table Styles, Forms and Form Styles.

Practical Demonstration of the followings

- 1) Display your family information with background and other formatting
- 2) Design a webpage to display your college with hyperlink.
- 3) Write an HTML documents to print the following:
 - a. Animal
 - i. Lion
 - ii. Tiger
 - iii. Elephant
 - b. Colours
 - i. Red
 - ii. Yellow
 - iii. Green
- 4) Write HTML document to illustrate the uses of the following tags with all attributes.
<H1>THROUGH<H6>,<P>,,<I>,<BLOCKQUOTE>,<PRE>,,,<DL>,<DT>,<DD>,<HR>,<U>,<SUB>,<SUP>,<STRIKE>,<CENTER>,,<MARQUEE>,,<A>
- 5) Develop and demonstrate a XHTML document that illustrates the use external style sheet, ordered list, table, borders, padding, color, and the tag.
- 6) Write HTML document to create top and dropdown navigation bar
- 7) Write a set of frames to show the following data

Fruits Animal Birds	Select any of the item shown on the left
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- 8) Write HTML document to create a given table structure using all the table related tags to include the following features:

One column	One row and three columns	Two rows and three columns
100	100 200 300	100 200 300 400 500 600

- Different colors and background Images
 - Spanning of rows and columns
 - Cell spacing and padding.
- 9) Write HTML document to create a given table structure using all the table related tags to include the following features:

Display the picture MIT	STUDENTS FEE RECORD			
	Roll No.	First Name	Last Name	Fees paid
Row 1	1			
Row 2	2			
Row 3	3			

- 10) Develop an HTML document for a web page of your favourite City. Design page with an attractive background color, text color and background image.

Recommended Books:

1. Lemay Colburn Kyrnin: *Mastering HTML, CSS & Javascript Web Publishing*, BPB Publications (2016)
2. Thomas. A. Powell: *The Complete HTML and CSS References*, 5th Edition, McGraw Hill (2010).
3. Jon Duckett: *HTML & CSS: Design and Build Web Sites*, John Wiley & Sons (2011)
4. Jon Duckett: *Javascript and JQuery: Interactive Front-end Web Development*, CreateSpace Independent Publishing Platform (2017)

Note: The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary

First Semester
Name of Paper: Office Automation Lab
Course No.: BCA/1/CC/06

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: **3**
(0- 0- 3)

A. Record Book	(10 marks)
B. Viva Voce	(15 marks)
C. Practical	(50 marks)

Unit-1: Microsoft Word 2010

Word Introduction, The Workplace, Typing and Editing, Bullets and Numbering, Finding and Replacing, AutoCorrect, Formatting Text and Paragraphs, Header and Footers, Sorting Lists, Working with Tables and Using Formulae in Tables, Graphics and Multimedia, Page Design and Layout, File Management, Opening, Saving, and Protecting Documents, Printing, Mail Merge.

Practical may be given for

1. Creating the documents with Special effects like font, color, underline, bold, different size.
2. Find and Replace operations like cut, paste, copy clipboard.
3. Bullets, Numbering and sorting
4. Paragraphs, alignment, indentation, line spacing
5. Inserting Pictures
6. Printing the documents, it includes paper-size, margins, header and footer, page no.
7. Creating a table, drawing table.
8. Using reference tools like table of contents, footnote, endnote
9. Using review tools- spell check, comments, document protection
10. Mail merge

Unit 2: Microsoft Excel 2010

Introduction to Excel, Working in Workbooks and Worksheet, Entering Data, Using Formulas and Functions to Calculate Values, Editing a Worksheet, formatting a Worksheet, Creating Charts, (graphs), Chart Types, Formatting Chart, Data sorting and filtering Data in a List, Adding Comments to Documents, Recording and Using Macros, Saving and Protecting a Workbook.

Practical may be given for

1. Creating Worksheets.
2. Formatting cells, Worksheets etc.
3. Printing, Inserting, Deleting, Copying, Moving worksheets.
4. Formulas, Built-in functions.
5. Graph-Plotting facilities.
6. Protection facility
7. Macro facility

Unit 3: PowerPoint 2010

PowerPoint Basics, Creating New Slide, Inserting and Deleting slides, Slides Layout, Inserting sound, picture and multimedia objects, creating different charts, Setting and Editing Slide Designs, Setting Transition Effects, Choosing Animations, Setting Custom Animations, Rehearsing and Viewing Slide Shows.

Unit 4: Access 2010

Access Basics, Creating Tables, Setting Primary Keys, Table Relationships, Creating Views, Using Operators in Queries, Creating Access Forms, Creating Reports.

Recommended Books:

1. The AGI Training Team: *Microsoft Office 2010 Digital Classroom*, Wiley Publishing Inc (2011).

Note: *The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*